

Comparing Time-Based and Event-Based Trading Rules on a Multi-Asset ETF Universe

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Project Understanding Summary

The uploaded project implements a compact research pipeline for comparing four portfolio rules on daily ETF data downloaded from Yahoo Finance: Buy & Hold, Time-Series Momentum, Directional Change (DC), and Multi-Threshold Directional Change. The tested asset universe is SPY, QQQ, TLT, and GLD, with data beginning on January 1, 2010 and extending through April 20, 2026.

The methodology is clear from the source code. Time-Series Momentum uses a 252-trading-day lookback return and assigns a signal of +1 when the trailing return is positive, -1 when it is negative, and 0 otherwise. The DC strategy is event-based: for each asset, the code tracks local highs and lows and flips trend state only when price moves by at least a fixed threshold of 2% from the most recent extreme. The Multi-Threshold DC extension combines three DC models with thresholds of 1%, 2%, and 3%, using weights of 0.5, 0.3, and 0.2 respectively; the aggregated weighted vote is then mapped back into a long or short signal.

All active strategies are backtested using one-day lagged signals to avoid look-ahead bias. Signals are normalized each day so that the sum of absolute active exposures equals one, which means the portfolio scales exposure only across the assets that are currently active. Daily turnover is measured as the absolute change in normalized portfolio weights, transaction costs are applied as turnover multiplied by a cost rate, and trade count is measured from signal changes at the asset level. Performance is summarized using total return, annualized Sharpe ratio, annualized volatility, maximum drawdown, average daily return, turnover, and number of trades. A robustness test reruns the active strategies under transaction cost rates of 0, 5, and 10 basis points.

The saved outputs show a very consistent pattern. Buy & Hold dominates in total return, Time-Series Momentum is the strongest active rule, and both DC variants underperform sharply once realistic trading frictions are included. The principal reason is visible in the turnover statistics: the DC strategies trade far more frequently, so transaction costs materially erode returns.

1 Abstract

This project evaluates whether event-based trading strategies can outperform a conventional time-based momentum rule on a diversified ETF universe. Using daily adjusted close data for SPY, QQQ, TLT, and GLD from Yahoo Finance over January 2010 to April 2026, the study compares Buy & Hold, Time-Series Momentum, a single-threshold Directional Change strategy, and a Multi-Threshold Directional Change extension. The backtesting framework applies lagged signals, normalized portfolio weights, and explicit transaction costs linked to turnover. The results show that Time-Series Momentum is the strongest active strategy, while both event-based Directional Change variants perform poorly after costs because their higher trading intensity generates substantial implementation drag. On this dataset and at daily frequency, the evidence does not support the claim that event-based strategies outperform time-based momentum.

2 Introduction

Algorithmic trading converts market information into systematic trading decisions through pre-defined rules. In quantitative finance, this is attractive because it replaces discretionary judgment with transparent and repeatable logic, allowing strategies to be tested across long samples and multiple assets (Chan, 2013). A key design choice in any algorithmic strategy is the timing rule that determines when the model updates its view of the market.

The most common approach is time-based. In a time-based strategy, signals are recalculated at fixed calendar intervals such as daily, weekly, or monthly. Time-Series Momentum is a classic example: the strategy examines past returns over a chosen lookback window and takes a long or short position depending on whether the asset has exhibited positive or negative trend behavior (Moskowitz et al., 2012).

An alternative is event-based trading. Rather than updating at every clock tick, event-based methods react only when price moves by a meaningful amount. Directional Change is a representative framework in this class. It defines market events through threshold moves from local extrema and treats confirmed reversals as new information (Guillaume et al., 1997; L’opez de Prado, 2018). In principle, this can reduce irrelevant noise and better align signals with market structure.

This project compares those two philosophies directly. The research question is:

Do event-based trading strategies outperform time-based momentum strategies on a multi-asset ETF dataset?

Answering this question is useful because event-based methods are often presented as more adaptive than fixed-interval models, yet their practical performance depends heavily on turnover, data frequency, and implementation costs. The present study therefore evaluates not only return profiles, but also drawdowns, trading activity, and cost sensitivity.

3 Data

The dataset consists of daily adjusted close prices downloaded from Yahoo Finance using the `yfinance` package (Yahoo, n.d.). The universe contains four liquid and widely used exchange-traded funds:

- SPY: broad US equity market exposure,
- QQQ: technology and growth-oriented US equities,
- TLT: long-duration US Treasury bonds,
- GLD: gold exposure.

The sample starts on January 1, 2010 and, because the project sets no fixed terminal date, extends to the most recent available observation in the saved output, namely April 20, 2026. After downloading adjusted close prices, the pipeline forward-fills missing observations, removes incomplete rows, and converts prices into simple daily returns.

This asset set is a reasonable teaching and research universe for three reasons. First, the ETFs are highly liquid and easy to interpret. Second, they cover distinct macro exposures, namely equities, technology-heavy equities, government bonds, and a commodity hedge. Third, using multiple assets allows the strategies to be evaluated in a portfolio setting rather than as isolated single-asset rules.

4 Methodology

4.1 Time-Series Momentum

The Time-Series Momentum model uses a 252-trading-day lookback, which is approximately one year. For each asset i at time t , the trailing return is computed as

$$R_{i,t}^{(252)} = \frac{P_{i,t}}{P_{i,t-252}} - 1.$$

The trading signal is then defined by the sign of this lookback return:

$$s_{i,t} = \begin{cases} 1 & \text{if } R_{i,t}^{(252)} > 0, \\ -1 & \text{if } R_{i,t}^{(252)} < 0, \\ 0 & \text{otherwise.} \end{cases}$$

This is a standard directional momentum rule. A positive trailing return is interpreted as continuation in trend and therefore generates a long signal, while a negative trailing return generates a short signal. Because the model uses a long horizon, it naturally trades less frequently than event-driven rules, which is consistent with the broader time-series momentum evidence in the literature (Jegadeesh & Titman, 1993; Moskowitz et al., 2012).

4.2 Directional Change (DC)

The Directional Change strategy is event-based rather than calendar-based. Instead of checking a fixed-horizon return, it monitors whether price has moved sufficiently far from a recent high or low to confirm a reversal. In the code, the threshold is set to 2%.

Operationally, the algorithm keeps track of the latest local high and local low. If the current trend is non-negative and price falls by at least 2% from the latest high, the state flips to a downtrend. If the current trend is non-positive and price rises by at least 2% from the latest low, the state flips to an uptrend. Between reversal events, the latest non-zero trend state is carried forward.

This construction is meant to capture market structure more directly than a fixed lookback rule. However, the quality of the signal depends critically on whether a 2% threshold filters noise effectively at the chosen sampling frequency, which is a central issue in event-based market modelling (Guillaume et al., 1997; L’opez de Prado, 2018).

4.3 Multi-Threshold DC

The Multi-Threshold DC strategy extends the single-threshold design by combining several event scales at once. In the project, three thresholds are used: 1%, 2%, and 3%, with weights 0.5, 0.3, and 0.2 respectively. For each threshold, a DC signal is produced independently. These signals are then combined through a weighted vote, and the final portfolio signal is set to +1 when the aggregate vote is positive and -1 when it is negative.

The rationale is that no single threshold is universally optimal. A smaller threshold reacts more quickly but may be more sensitive to noise, while a larger threshold reacts more slowly but may be more stable. By combining thresholds, the strategy aims to improve robustness and reduce dependence on one arbitrary parameter choice, an objective aligned with the broader event-based sampling perspective in financial machine learning (L’opez de Prado, 2018).

4.4 Backtesting Framework

The backtesting framework is implemented consistently across all active strategies and follows standard principles of systematic strategy evaluation (Pardo, 2008).

Signal lagging and look-ahead control. Signals are shifted by one trading day before being applied to returns. This means that day- t portfolio returns use only information known at day $t - 1$, preventing look-ahead bias.

Portfolio construction. The strategy takes the lagged signals and normalizes them each day so that the sum of absolute active weights equals one. If several assets are active, capital is distributed evenly in absolute terms across them. This creates a comparable portfolio scale across strategies even when the number of active positions changes over time.

Benchmark. Buy & Hold is implemented as an equal-capital benchmark across the four ETFs, with weights allowed to drift through time. Because the code does not rebalance this benchmark, it incurs no transaction costs.

Transaction costs and turnover. Daily turnover is defined as the sum of absolute changes in normalized portfolio weights from one day to the next. Net returns are computed as gross portfolio return minus turnover multiplied by the transaction cost rate. The baseline cost assumption is 5 basis points per unit of turnover, and robustness is examined at 0, 5, and 10 basis points.

Performance metrics. The report uses total return, annualized Sharpe ratio, annualized volatility, maximum drawdown, average daily return, turnover, and number of trades or position changes. Sharpe ratio measures risk-adjusted return, volatility captures return dispersion, drawdown quantifies peak-to-trough loss, and turnover and trade count measure implementation intensity.

5 Results

Table 1 reports the main performance statistics from the saved output files.

Table 1: Performance summary

Strategy	Total Ret.	Sharpe	Vol.	Max DD	Avg Daily	Turnover	Trades
Buy & Hold	649.70%	1.01	13.17%	-29.65%	0.05%	0.00%	0
Time-Series Momentum	76.62%	0.41	9.78%	-27.36%	0.02%	3.73%	308
Directional Change	-64.92%	-0.54	10.90%	-69.27%	-0.02%	18.85%	1540
Multi-Threshold DC	-70.59%	-0.63	10.94%	-74.50%	-0.03%	22.56%	1849

The performance table supports a more analytical interpretation than a simple ranking. Buy & Hold dominates with a 649.70% total return and a 1.01 Sharpe ratio, implying that the sample period rewards persistent exposure to the ETF basket more than frequent tactical reallocation. Among the active rules, Time-Series Momentum is the only strategy that converts its signal into economically meaningful positive net performance. It records a 76.62% total return, a 0.41 Sharpe ratio, 9.78% annualized volatility, and a maximum drawdown of -27.36%. By contrast, both DC variants produce negative total returns and negative Sharpe ratios despite volatility only modestly above momentum. Their weakness therefore comes primarily from poor return generation rather than from taking dramatically more risk.

The contrast between momentum and Directional Change is especially clear in the path of returns. In Figure 1, the DC equity curves do not fail in a single short episode; they drift downward for much of the sample. That pattern matters because it indicates persistent signal inefficiency rather than one isolated adverse regime. Figure 2 shows the same mechanism from a risk perspective: the DC strategies remain in deep drawdowns for extended periods, which means losses are not quickly reversed by later gains. Momentum, although far weaker than Buy & Hold, exhibits materially shallower and less persistent underwater periods, suggesting that it captures some durable trend exposure rather than repeatedly reacting to noise.

The drawdown chart in Figure 2 reinforces the same message. Time-Series Momentum experiences meaningful losses, but its drawdown profile remains manageable relative to the event-based models. The DC strategies suffer much deeper and more persistent underwater periods, which implies that their weak returns are accompanied by poor capital preservation. A strategy that loses money while also requiring investors to endure 69–75% drawdowns is difficult to justify in practice.

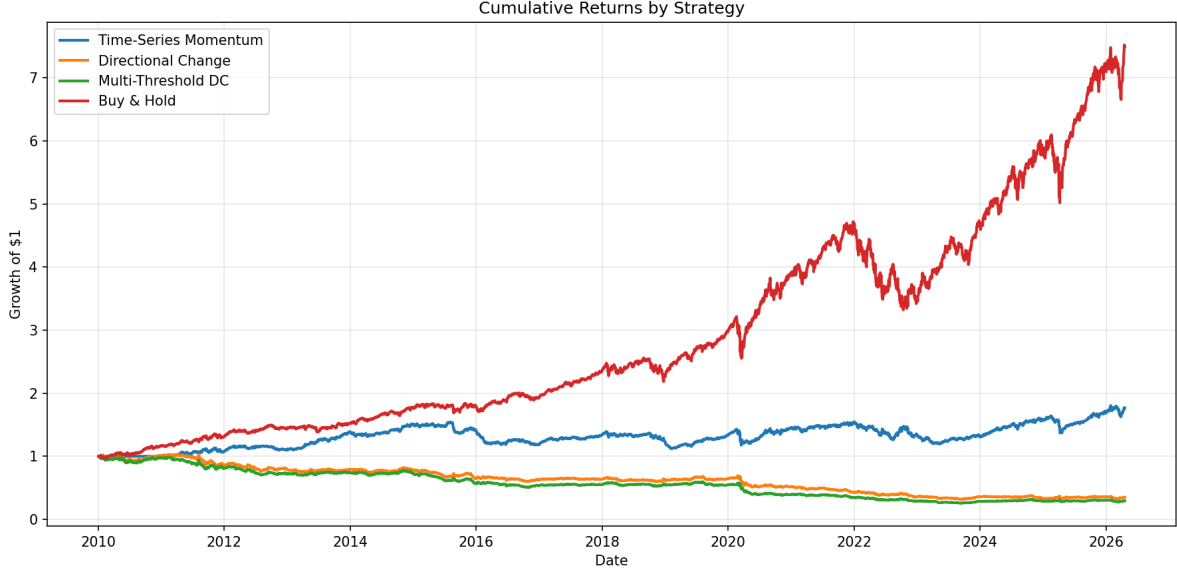


Figure 1: Cumulative returns by strategy.

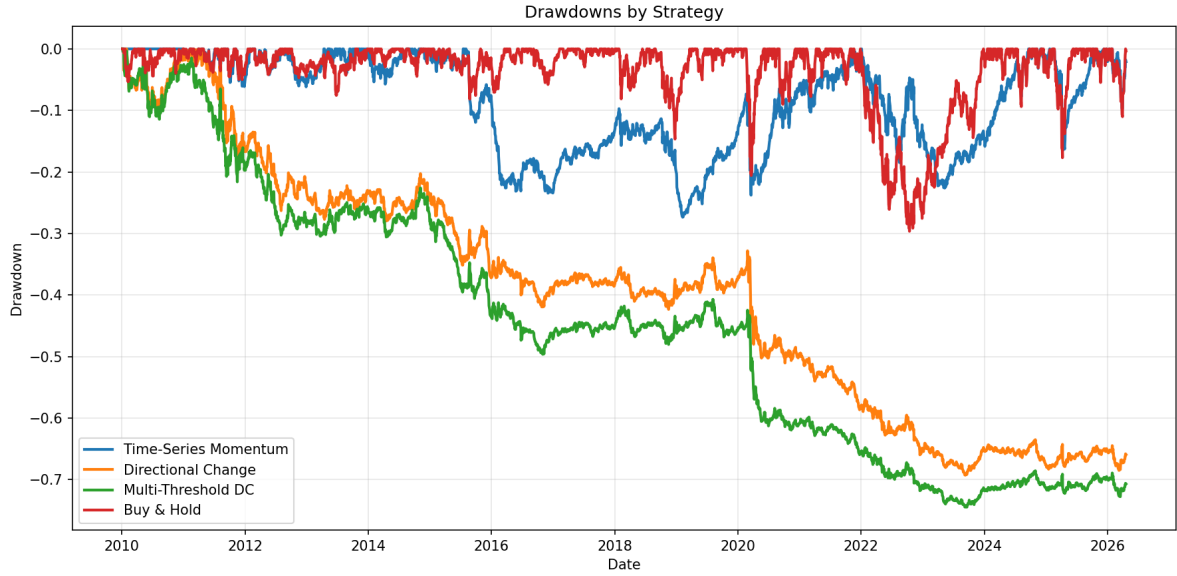


Figure 2: Drawdowns by strategy.

Turnover and trade count explain much of this underperformance. Table 2 shows that Time-Series Momentum has average turnover of 3.73% and 308 asset-level position changes, whereas Directional Change has 18.85% turnover and 1,540 position changes. The Multi-Threshold DC strategy is even more active, with 22.56% turnover and 1,849 position changes. In other words, the event-based rules trade roughly five to six times more intensely than the momentum model.

This trading intensity matters because transaction costs directly scale with turnover in the backtest. The cost sensitivity analysis in Table 3 confirms that the ranking is not fragile. Time-Series Momentum remains profitable across all tested cost assumptions, although its total return declines

Table 2: Turnover and trade count

Strategy	Turnover	Number of Trades
Buy & Hold	0.00%	0
Time-Series Momentum	3.73%	308
Directional Change	18.85%	1540
Multi-Threshold DC	22.56%	1849

from 90.68% at zero cost to 63.60% at 10 basis points. The DC strategies are already loss-making even before costs: Directional Change returns -48.37% at zero cost, while Multi-Threshold DC returns -53.28%. Once a 5 basis point cost is applied, losses deepen sharply, and at 10 basis points the total returns fall to -76.17% and -81.48% respectively.

Table 3: Transaction cost sensitivity

Strategy	Cost (bps)	Total Return	Sharpe	Turnover
Time-Series Momentum	0	90.68%	0.46	3.73%
Time-Series Momentum	5	76.62%	0.41	3.73%
Time-Series Momentum	10	63.60%	0.36	3.73%
Directional Change	0	-48.37%	-0.32	18.85%
Directional Change	5	-64.92%	-0.54	18.85%
Directional Change	10	-76.17%	-0.75	18.85%
Multi-Threshold DC	0	-53.28%	-0.37	22.56%
Multi-Threshold DC	5	-70.59%	-0.63	22.56%
Multi-Threshold DC	10	-81.48%	-0.89	22.56%

The key interpretation is that transaction costs do not merely reduce already strong DC profits; instead, they expose that the event-based rules are fundamentally too noisy on this dataset. Even with zero transaction costs, the DC models fail to generate attractive risk-adjusted returns. Costs then worsen an already weak gross signal. By contrast, Time-Series Momentum appears to have a genuine underlying edge on this sample, because it remains positive under every tested cost level.

6 Discussion

The evidence suggests that Time-Series Momentum outperforms the event-based alternatives because it is better aligned with the characteristics of daily ETF data. A 252-day lookback is slow-moving and therefore captures broad, persistent trends rather than short-lived fluctuations. On a diversified ETF universe that includes strong secular equity trends, this type of signal is naturally more stable and incurs less trading friction.

The DC strategies likely struggle for three related reasons. First, daily data may be too coarse for threshold-based reversal detection to provide a real informational advantage. Directional Change methods are often more compelling when applied to higher-frequency data, where event sampling can meaningfully filter microstructure noise and adapt to uneven information arrival. On daily bars, however, a 1–3% threshold may still react to ordinary market variation rather than economically meaningful reversals.

Second, the turnover statistics show that the DC rules are implementation-heavy. Frequent state flips force repeated portfolio reallocation, and the backtest penalizes that behavior through transaction costs. The results show that this is not a minor effect: the cost drag is large enough to materially deteriorate already weak performance.

Third, the multi-threshold extension does not improve outcomes. In principle, combining thresholds should make the signal more robust by reducing dependence on one parameter choice. In this sample, however, the weighted voting mechanism appears to amplify trading activity rather than improve trend detection. The strategy becomes even more active than the single-threshold DC rule and therefore suffers even more from cost drag.

6.1 When Might Event-Based Strategies Work?

Directional Change may be better suited to settings in which price discovery occurs through many small, irregular moves rather than through relatively smooth daily trends. At higher sampling frequencies, event-based sampling can be more informative because the price path contains richer local structure, allowing reversal thresholds to react to genuinely meaningful changes rather than to broad end-of-day averages. The approach may also be more effective in more volatile or mean-reverting assets, such as foreign exchange, commodities, or cryptoassets, where short-horizon reversals are more common than in the diversified ETF universe studied here. In those environments, a threshold rule can function as a state-dependent filter, updating the signal only when price movement is large enough to be economically informative. Its success, however, depends heavily on threshold selection. Thresholds that are too small are likely to react to noise and recreate the whipsaw trading documented in this paper, whereas larger thresholds may improve signal quality by filtering out minor fluctuations. Even then, the strategy remains most plausible in low-cost trading environments, because any advantage from faster event detection can be offset quickly if frequent rebalancing is accompanied by meaningful fees, spreads, or slippage.

Several limitations should be acknowledged. The first is data frequency. Because the analysis uses daily data, it may understate the potential value of event-based techniques designed for more granular market dynamics. The second is the small asset universe: four ETFs are sufficient for a clean classroom project, but not for drawing broad conclusions about all asset classes or all market regimes. The third is the simplicity of the backtest. The framework models proportional transaction costs through turnover, but it does not include slippage, financing costs, short-sale constraints, or more complex portfolio risk controls. These assumptions are acceptable for comparative research, but they still simplify real implementation.

7 Conclusion

The answer to the research question is no: on this multi-asset ETF dataset, the event-based Directional Change strategies do not outperform the time-based momentum strategy. Time-Series Momentum is the strongest active approach, producing positive returns, a positive Sharpe ratio, lower turnover, and materially shallower drawdowns than either DC variant.

More broadly, the results show that an event-based idea is not automatically superior simply because it appears more adaptive. On daily ETF data, the Directional Change models trade too

frequently, absorb too much noise, and suffer substantial cost drag, whereas the slower momentum rule proves more robust and more implementable. Their effectiveness is therefore likely to depend on data frequency, asset characteristics, threshold calibration, and the cost of implementation. This conclusion is consistent with the broader literature showing that signal design and implementation discipline matter as much as conceptual elegance (Chan, 2013; Pardo, 2008).

8 References

Core Methodology References

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Data Source

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